# LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

**Generate a Research Paper with AI, Then Audit Its Citations**

*AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40*

| Student Detail | Fill in |
|---|---|
| Name | AMAN BHAGAT |
| Roll Number | MCL2026013 |
| Programme / Department | IT (CLIS) |
| Topic ID | T3_____ |
| Full Assigned Topic | Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models |
| AI Tool Used | Gemini |
| Date | 21/08/2026 |

**IMPORTANT:** You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

## 1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

**Question 1 - Citation Authenticity:** Does the cited publication actually exist, and is it correctly identified?

**Question 2 - Claim-Citation Support:** If the publication is genuine, does it actually support the claim for which the AI cited it?

## 2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

**Most important principle:** You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

## 3. WHAT YOU WILL DO

| Part | Activity |
|---|---|
| A | Record your baseline perception of AI citations |

| | |
|---|---|
| B | Record the AI environment used |
| C | Generate and preserve an AI-written research paper |
| D | Create a reference inventory |
| E | Select 10 references systematically |
| F | Judge their plausibility before verification |
| G | Verify authenticity, metadata and identifiers |
| H | Determine whether genuine references support the AI-generated claims |

## 4. RECOMMENDED 2-HOUR TIME PLAN

| Time | Activity |
|---|---|
| 0-10 min | Parts A-B: Baseline and AI environment |
| 10-30 min | Part C: Generate and save research paper |
| 30-40 min | Parts D-E: Inventory and select references |
| 40-45 min | Part F: Pre-verification plausibility |
| 45-80 min | Part G: Authenticity and metadata audit |
| 80-110 min | Part H: Claim-citation entailment audit |
| 110-120 min | Calculate scores, check worksheet and prepare submission |

**Do not spend excessive time formatting the AI-generated paper.** The audit is the main activity.

## PART A - BASELINE ASSESSMENT

**Complete this section BEFORE opening the AI tool.**

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☑ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☑Once or twice ☐Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☑ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☑ 2 ☐ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

**Important:** Do not change this answer later, even if your opinion changes during the experiment.

## PART B - RECORD YOUR AI ENVIRONMENT

| Information | Your Response |
|---|---|
| AI tool used | Gemini |
| Model name | Pro |
| Model/version shown | 3.1 |
| Access tier | Institutional |
| Web browsing available? | Yes |
| Was web browsing actually used? | Yes |
| Research/Deep Research mode used? | No |
| Date/time generated | 10 AM 21/08/2026 |

**Important:** If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

# PART C - GENERATE THE AI RESEARCH PAPER

## C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

## C2. Use the Following Common Prompt

**Replace only [**Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models**].**

> *Generate a scholarly research paper on the topic: "[*Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models*]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

## C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

**STOP:** Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

# PART D - PAPER PROFILE AND REFERENCE INVENTORY

**Paper title:** Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models

| Measure | Value |
|---|---|
| Approximate pages | 4 |
| Approximate word count | 2176 |
| Number of sections | 14 |
| Total number of references | 4 |
| Approximate in-text citation occurrences | 19 |
| Did AI warn you to verify references? | No |

## D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

| Reference Type | Count |
|---|---|
| DOI available | 3 |
| PMID available, no DOI | 0 |

| | |
|---|---|
| arXiv ID available, no DOI/PMID | 0 |
| URL only | 0 |
| No persistent identifier supplied | 1 |
| TOTAL | 4 |

**Examples**

| Category | Example | How to Count |
|---|---|---|
| DOI available | Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730 | DOI available |
| PMID available, no DOI | A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found | PMID available, no DOI |
| arXiv ID available | Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629 | arXiv ID available |
| URL only | World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID | URL only |
| No persistent identifier | Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. | No persistent identifier if none is supplied |

**Important:** Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

## PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

**If the bibliography contains 10 or fewer references:** audit all references.

**If it contains more than 10 references:** select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

| Audit Slot | Original Bibliography Reference No. |
|---|---|
| R1 | 10.18653/v1/2024.findings-acl.558 |
| R2 | 2406.15927 |
| R3 | 2505.08200 |
| R4 | n.d. |
| R5 | |
| R6 | |
| R7 | |
| R8 | |
| R9 | |
| R10 | |

**Important:** From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

## PART F - PLAUSIBILITY TEST

**Do this BEFORE searching for any reference.**

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

| Score | Meaning |
|---|---|
| 1 | Obviously suspicious |
| 2 | Somewhat suspicious |
| 3 | Unsure |
| 4 | Looks genuine |
| 5 | Completely convincing |

| Ref. | Initial Impression | Plausibility 1-5 | Predict Genuine? |
|---|---|---|---|
| R1 | | 5 | Yes |
| R2 | | 4 | Yes |
| R3 | | 4 | Yes |
| R4 | | 3 | No |
| R5 | | | Yes / No |
| R6 | | | Yes / No |
| R7 | | | Yes / No |
| R8 | | | Yes / No |
| R9 | | | Yes / No |
| R10 | | | Yes / No |

**Example:** A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

**Critical rule:** Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

# PART G - CITATION AUTHENTICITY AND METADATA AUDIT

## G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

## G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

**Not acceptable as sole evidence:** "I asked another AI and it said the paper exists." AI can help you search, but final verification must rely on scholarly or publisher evidence.

## G3. Assign One A-E Code

| Code | Classification | Meaning |
|---|---|---|
| A | VERIFIED | Publication exists and important metadata are correct. |
| B | WRONG METADATA | Publication exists, but one or more bibliographic details are incorrect. |
| C | FRANKENSTEIN | Real-looking pieces are combined, but that specific publication does not exist. |
| D | FABRICATED | No reliable scholarly evidence that the publication exists. |
| E | IDENTIFIER MISMATCH | DOI/arXiv/PMID is invalid or resolves to a different publication. |

## G4. Examples

**A:** AI gives a paper and all details match the publisher record.

**B:** The paper exists, but the AI gives the wrong publication year or journal.

**C:** The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

**D:** You cannot find the paper or a credible scholarly record after systematic searching.

**E:** The DOI supplied by AI opens an entirely different paper or does not resolve.

## G5. Record Your Evidence

| Ref. | Publication Found? | Title Match? | Authors Match? | Year Match? | Venue Match? | Identifier Match? |
|---|---|---|---|---|---|---|
| R1 | Y | Y | Y | Y | Y | Y |
| R2 | Y | Y | Y | Y | Y | Y |
| R3 | Y | Y | Y | Y | Y | Y |
| R4 | Y | Y | N | N | N | N/A |
| R5 | | | | | | |
| R6 | | | | | | |
| R7 | | | | | | |
| R8 | | | | | | |
| R9 | | | | | | |
| R10 | | | | | | |

| Ref. | Verification Source(s) | Identifier Checked | What You Found |
|---|---|---|---|
| R1 | ACL Anthology | DOI: 10.18653/v1/2024.findings-acl.558 | Publication exists and the title, authors, year, venue and DOI match. |
| R2 | arXiv and DBLP | arXiv: 2406.15927 | The arXiv record identifies the paper as *Semantic Entropy Probes: Robust and Cheap Hallucination Detection in LLMs* by Jannik Kossen and co-authors, submitted in 2024 under arXiv ID 2406.15927 |
| R3 | arXiv | arXiv:2505.08200 | A comparison with the official record showed that the title, complete author list, publication year, and identifier were all accurately reported. |
| R4 | arXiv | No identifier was provided by the AI. The publication was located through an exact-title search, correct arXiv ID: 2605.28264. | An exact-title search confirmed that the publication is genuine, but the AI reference omitted four co-authors, used 'n.d.' instead of 2026, and did not provide the arXiv identifier |
| R5 | | | |
| R6 | | | |
| R7 | | | |
| R8 | | | |

| | | | |
|---|---|---|---|
| R9 | | | |
| R10 | | | |

## G6. Final Classification

| Ref. | Final Code | One-Line Reason |
|---|---|---|
| R1 | A | The paper exists, and the title, authors, year, venue, and DOI all matched the official record. |
| R2 | A | The paper was verified in the official arXiv record, and all major bibliographic details, including the authors, year, title, and identifier, were found to be accurate. |
| R3 | A | The publication was verified, with all major bibliographic details and the arXiv identifier found to be accurate. |
| R4 | B | The publication exists, but the AI citation contains incomplete authorship, an incorrect year, and inaccurate source details. |
| R5 | | |
| R6 | | |
| R7 | | |
| R8 | | |
| R9 | | |
| R10 | | |

**Good reason:** B: Article exists, but AI reported 2022 instead of 2020.

**Poor reason:** B: Details wrong.

## PART G - SCORING

Count your final classifications: A = 3 B = 1 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 4

| Code | Points |
|---|---|
| A | 4 |
| B | 3 |
| C | 1 |
| D | 0 |
| E | 1 |

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(3) + 3(1) + 1(0) + 0(0) + 1(0) = 15 points

Authenticity Score = 15 / (4 x 4) x 100 = 93.75 /100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

| Score | Interpretation |
|---|---|
| 90-100 | Excellent authenticity |
| 75-89 | Good, minor concerns |
| 60-74 | Moderate citation-authenticity risk |
| 40-59 | High citation-authenticity risk |
| Below 40 | Very high citation-authenticity risk |

## G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 4   Total predictions = 4   Prediction Accuracy = 100 %

## FINAL LAB 1 RESULTS

| Measure | Your Result |
|---|---|
| Total references in AI paper | 4 |
| References deeply audited | 4 |
| A: Verified | 3 |
| B: Wrong metadata | 1 |
| C: Frankenstein | 0 |
| D: Fabricated | 0 |
| E: Identifier mismatch | 0 |
| Authenticity Score | 93.75 /100 |
| Prediction Accuracy | 100 % |

## FINAL INTERPRETATION

**1. Did professional-looking citations necessarily turn out to be genuine?**

No. Although most of the citations were verified as genuine, one citation contained incorrect and incomplete bibliographic details. This shows that a professional-looking citation should still be verified before being trusted.

**2. Did all genuine references actually support the claims for which they were cited?**

Yes. After checking the original sources, all four references were found to support the main claims made about them in the AI-generated paper. However, R4 showed that a source can support a claim even when R4 had metadata problems.

**3. What was the most serious citation-integrity problem you observed?**

The most serious problem was incorrect bibliographic metadata in R4, where important details such as the complete author list, publication year, source information, and identifier were missing.

**4. What is the single most important lesson you learned from this lab?**

The most important lesson I learned is that a citation should not be trusted simply because it looks genuine; both the existence of the source and its support for the cited claim must be verified.

## SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

## STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

**Student Name:** Aman Bhagat

**Roll Number:** MCL2026013

**Signature:** Aman

**Date:** 21/08/2026

## CENTRAL LESSON OF LAB 1

### Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission